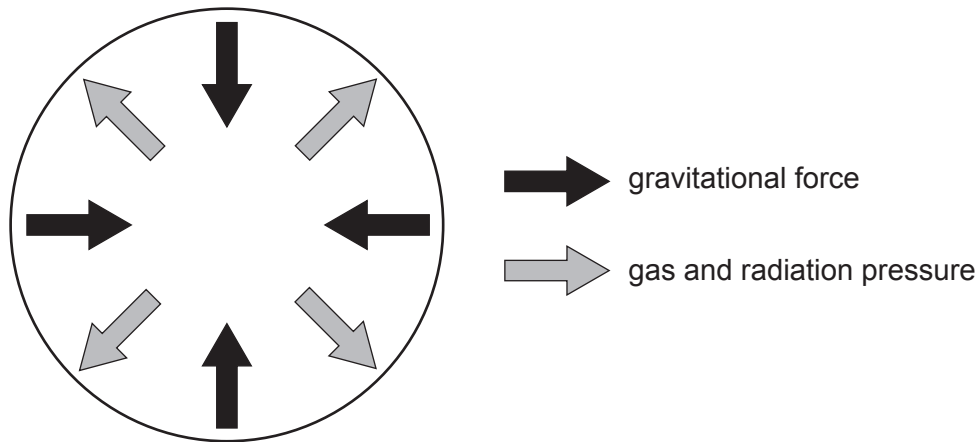


2. The diagram below shows the forces acting on a high-mass star that is on the main sequence.



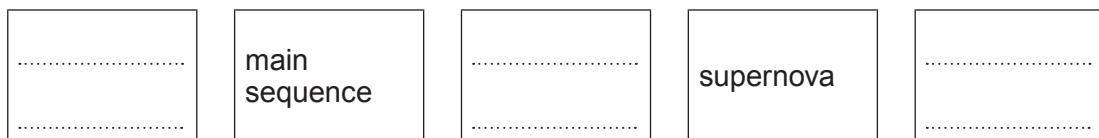
(a) Complete the sentences below by circling the correct answer.

(i) The star is (**burning** / fusing / melting) hydrogen. [1]

(ii) The star is currently stable because the forces acting on it are (**balanced** / unbalanced / zero). [1]

(b) (i) Use the words from the box to complete the diagram to show the stages in the life cycle of a **high-mass** star. [3]

supergiant protostar yellow dwarf red giant neutron star white dwarf



(ii) Tick (✓) the box next to the correct statement to complete the following sentence. [1]

During the supernova stage of the star's life cycle:

the star is red shifted

☐

heavy elements are returned to space

☐

nuclear fission takes place

☐
